No.	Status	Date
LNDE-RPT-20260803-034-DS-2-RF-102	Issued	2026-08-03

UT Thickness Data Sheet - 2-RF-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 2-RF-102 located within the ATU-100. The scope of work involves establishing a new-spool baseline for replacement spools installed per MOC, as detailed in LNDE-RPT-20260803-034-R0.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the thickness data.

Technical Basis and Procedure

Thickness measurements were performed under QSCP-NDT-001 Rev 4. The inspection was conducted in accordance with the requirements of ASTM E797.

As the line is constructed from A312 TP316, all measurements were taken using the stainless velocity setting on LNDE-SS-003 to ensure accuracy for the 304L CMLs.

Line Design Parameters

The subject line, 2-RF-102, is a Reflux line with a nominal outer diameter of 60.3 mm and a nominal wall thickness of 3.91 mm. The design pressure is 517 kPa(g) and the design temperature is 104 °C.

The minimum allowable thickness (t-min) for this line is 2.16 mm. Any reading recorded below this threshold is subject to an NCR.

Inspection Results

The baseline thickness data for the replacement spools on 2-RF-102 has been captured. These readings serve as the initial data point for future corrosion rate calculations and remaining life assessments.

The measured values are recorded in the associated data tables, confirming the current state of the A312 TP316 material relative to the t-min of 2.16 mm.

Thickness readings, measured 2026-08-03

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2025-08-15	6.32	6.23	0.075	0.155	15.6	2029-08-03	
4-RA-101-02	8.56	3.81	2025-08-15	6.40	6.23	0.076	0.144	16.8	2029-08-03	
4-RA-101-03	8.56	3.81	2025-08-15	5.42	5.12	0.115	0.222	5.9	2029-07-17	gauge photo attached
4-RA-101-04	8.56	3.81	2025-08-15	6.82	6.65	0.065	0.144	19.7	2029-08-03	
4-RA-101-05	8.56	3.81	2025-08-15	6.20	6.05	0.084	0.175	12.8	2029-08-03	
4-RA-101-06	8.56	3.81	—	—	8.56	—	—	—	2027-02-01	
4-RA-101-07	8.56	3.81	2025-08-15	7.10	7.07	0.050	0.111	29.4	2029-08-03	
4-RA-101-08	8.56	3.81	2025-08-15	4.85	4.54	0.136	0.260	2.8	2027-12-27	gauge photo attached
4-RA-101-09	8.56	3.81	2025-08-15	7.04	6.97	0.053	0.103	30.6	2029-08-03	
4-RA-101-10	8.56	3.81	2025-08-15	6.86	6.57	0.069	0.160	17.3	2029-08-03	
4-RA-101-11	8.56	3.81	2025-08-15	5.34	5.18	0.115	0.229	6.0	2029-07-29	gauge photo attached
4-RA-101-12	8.56	3.81	2025-08-15	5.95	5.77	0.185	0.165	10.6	2029-08-03	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2025-11-03	3.52	3.59	0.011	0.010	>50	2032-08-03	
2-RF-102-02	3.91	2.16	2025-11-03	3.49	3.49	0.013	0.024	>50	2032-08-03	
2-RF-102-03	3.91	2.16	2025-11-03	3.64	3.66	0.009	0.003	>50	2032-08-03	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2027-08-03	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-05	3.91	2.16	—	—	3.90	—	—	—	2027-08-03	
2-RF-102-06	3.91	2.16	—	—	3.96	—	—	—	2027-08-03	
2-RF-102-07	3.91	2.16	—	—	3.87	—	—	—	2027-08-03	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2026-08-03	_____